
TATA STEEL AI HACKATHON — ROUND 2 CHALLENGE SUBMISSION

Maintenance Wizard

AI-Powered Condition-Based Predictive Maintenance Decision-Support Platform

Executive Summary & Interview Guide: The Maintenance Wizard is an advanced AI-driven decision-support system designed for heavy manufacturing steel plants. By orchestrating a **6-agent reasoning brain** (powered by Google Gemini and Llama 3.3), the platform processes real-time telemetry sensor records (vibration, temp, pressure) to detect anomalies (using scikit-learn Isolation Forests), predict Remaining Useful Life (RUL), cross-reference warehouse inventory levels for spare parts optimization, and perform semantic search over standard operating procedures (SOPs) using local ChromaDB RAG. This 100-page comprehensive manual serves as a complete technical guide, covering multi-agent specifications, prompt layouts, FMEA matrices, API schemas, installation steps, and individual profiles for 30 critical assets.

Presenter: Shubham Chakrawarti, Lead Platform Architect & Developer

Project Repository: github.com/shubha9696/maintenance-wizard

Live Production App: ai-maintenance-wizard.vercel.app

Live Backend API: maintenance-wizard-backend.onrender.com/docs

Document Specifications: Complete 100-Page System Dossier & Technical Reference Manual

Date: June 2026

1. Executive Summary & Project Abstract

Industrial manufacturing environments, particularly modern steel plants, are defined by high-throughput, continuous-process operations where unexpected failure of a single critical machine can stall downstream production flow. Historically, maintenance in these plants has followed a mix of manual interval-based inspection schedules and reactive repair runs. While preventative maintenance schedules provide a baseline, they are financially inefficient—components are frequently replaced before the end of their functional lifespan, and sudden breakdown incidents still occur.

The **Maintenance Wizard** represents an operational paradigm shift. By using condition-based telemetry stream processing, predictive anomaly modeling, and retrieval-augmented agentic coordination, the platform gives control room operators and maintenance technicians a conversational decision support console.

The system acts as a digital twin brain, mapping sensor signals directly to structural degradation curves, cross-checking warehouse spares availability, and auto-generating safety-compliant work cards.

2. Industrial Plant Context & Operations

Within integrated steel plants, operations are distributed across multi-stage facilities: Blast Furnaces, Steel Melting Shops (SMS), Rolling Mills, Coke Ovens, Sinter Plants, and Power Utilities. Each sector operates under harsh environmental conditions characterized by extreme heat, high pressure, and abrasive dust particles.

These settings house a massive fleet of high-stress components:

- Centrifugal pumps that circulate water jackets surrounding furnace shells.
- Large converter vessels tilting hundreds of tons of liquid pig iron.
- High-speed rolling motors driving slab reduction lines.

Because these assets are historically maintained by different engineering teams, telemetry data and maintenance logs remain heavily siloed. A diagnostic engineer checking vibration signatures lacks visibility into recent mechanical torque alerts, spare parts replenishment orders, or compliance checklists, leading to extended mean time to repair (MTTR).

3. Problem Statement: The Cost of Information Silos

Unplanned downtime in heavy manufacturing carries severe direct and indirect financial penalties:

- **Direct Financial Losses:** Stopping a blast furnace or continuous casting line halts iron flow, costing up to \$10,000 to \$25,000 per hour in lost output.
- **Component Collateral Damage:** Running a motor with a degraded bearing can damage the rotor, turning a simple bearing swap into an expensive spindle rebuild.
- **Information Latency:** When an alarm fires, technicians must locate paper manuals, search local files for previous incident records, and manually verify warehouse part balances. This latency delays immediate shutdown alerts and puts technicians in high-risk situations.

Therefore, there is a critical need for an integrated platform that fuses real-time telemetry anomaly classification with automated information retrieval.

4. Fused Agentic Coordination Concept

The core concept of the Maintenance Wizard is **Multi-Agent Collaborative Orchestration**. Instead of employing a single, general-purpose LLM, the architecture partitions cognitive tasks among specialized agents that coordinate to solve complex queries.

For example, when a user asks: *'Pump BF-CP-001 has high vibration. Can I replace it tonight and are the parts available?'* the orchestrator coordinates three distinct agents:

1. The **Diagnostic Agent** retrieves and evaluates the sensor telemetry.
2. The **Knowledge Agent** runs a RAG search over manuals to find the replacement procedure.
3. The **Recommendation Agent** queries the spare parts database to verify inventory availability, calculate lead times, and outline lock-out tag-out (LOTO) protocols.

The results are synthesized into a single response, including the agents' thought processes.

5. Frontend Design System & Holographic UI

The client interface is designed as a Next.js 14 Web Application built on a custom space-dark design system:

- **Interactive Digital Twin:** Displays a spatial view of the plant floor with color-coded nodes indicating real-time status (Green: Operational, Yellow: Degraded, Red: Critical).
- **3D Thermal Stack:** Implements a rotating 3D isometric stack visualizer to display thermal and pressure profiles inside the blast furnace shell.
- **Analytics & ROI Panels:** Renders interactive degradation charts (using Recharts), fleet-wide health status histograms, and financial value metrics.
- **Real-Time Thought Consoles:** Embedded terminal displays next to the chat window show the orchestrator's reasoning steps, including classification scores, search terms, and threshold checks.

6. Backend Framework & Telemetry Handling

The backend is built on FastAPI (Python 3.10), chosen for its asynchronous capability and speed:

- **Lifespan Handlers:** On startup, the server bootstraps synthetic JSON databases (equipment, logs, spare parts) if missing, loads pre-trained anomaly models, and runs vector store checks. This ensures uvicorn begins serving requests within milliseconds.
- **CORS Middleware:** Configured to allow cross-origin requests, letting the hosted Next.js frontend communicate with the Render API server.
- **API Router Architecture:** Endpoints are partitioned into distinct routers:
 - /api/chat: Message endpoint routing to the agentic core.
 - /api/equipment: Asset metadata and dashboard stats.
 - /api/alerts: Incident logs and acknowledges.
 - /api/reports: Diagnostic report compiles.
 - /api/knowledge: Dynamic RAG file uploads.

7. Orchestrator Agent Specifications

The **Orchestrator Agent** is the routing brain of the platform. It handles incoming queries by executing the following pipeline:

- **Intent Classification:** Uses a zero-shot prompt layout to categorize user intent into specialized agent domains (incident_diagnostics, predictive_rul, spares_recommendations, custom_report, knowledge_rag, general_chat).
- **Entity Resolution:** Parses queries to identify specific equipment nodes (e.g. BF-CP-001) or plant sectors (e.g. Steel Melting Shop).
- **Context Propagation:** Maintains session history using conversational buffers to carry over context (such as equipment ID) across multi-turn exchanges.

The orchestrator enforces a clean separation of concerns, calling downstream agents asynchronously.

8. Diagnostic & Prediction Agent Implementations

These agents focus on telemetry data processing:

- **Diagnostic Agent:** Evaluates live sensor telemetry against high/low limits. When telemetry (like vibration) exceeds the normal threshold, it diagnoses potential failure modes (e.g. Cavitation, Overheating, or Unbalance) and outputs severity scores.
- **Prediction Agent:** Uses an Isolation Forest algorithm from scikit-learn. It evaluates multi-variate telemetry grids (vibration, heat, pressure, current) to calculate an overall health score (0 to 100) and remaining useful life (RUL). It identifies anomalies by assessing how far a machine's data deviates from normal parameters.

9. Recommendation & Report Agent Implementations

These agents focus on actionable outcomes:

- **Recommendation Agent:** Generates step-by-step repair guides. It cross-references the diagnostics with plant safety compliance rules and available spare parts, outputting safety precautions (LOTO) and listing the exact part numbers needed.
- **Report Agent:** Fuses data from diagnostic and prediction modules to compile formatted markdown reports, including Maintenance Summaries, Incident Reports, and Equipment Health sheets.

10. Knowledge Agent & Vector Database Schema

The **Knowledge Agent** handles unstructured data. It manages a persistent ChromaDB instance with 4 distinct vector collections:

- 1. **Collection knowledge_docs:** Stores parsed manuals, standard operating procedures, and safety compliance guides.
- 2. **Collection maintenance_logs:** Stores historical maintenance logs, filtered to focus on valuable repairs and corrective actions.
- 3. **Collection failure_reports:** Stores post-incident analysis reports, detailing symptoms, root causes, and lessons learned.
- 4. **Collection failure_modes:** Stores FMEA (Failure Mode and Effects Analysis) databases mapping equipment types to root causes.

11. RAG Ingestion Pipeline & Text Processing

The RAG (Retrieval-Augmented Generation) ingestion pipeline handles dynamic document uploads:

- **Document Parsing:** Dynamically parses incoming files (PDF, MD, TXT, JSON, CSV) based on file type. PDFs are processed using the pypdf library.
- **Text Chunking:** Splits raw text into chunks of 800 characters with a 100-character overlap. This balance maintains context across chunks while keeping them concise for embedding.
- **Vector Generation:** Generates vector embeddings for each chunk using the models/gemini-embedding-2 model, which is configured to output 3072-dimensional vectors.

12. Verification: Local RAG Ingestion logs

We verified the ingestion flow locally. Below are the execution logs from uploading a custom, dynamically compiled standard operating procedure PDF:

```
[INFO] Initializing dynamic file parser for type: SOP...
[INFO] Preparing file payload: test_blast_furnace_sop.pdf (0.01 MB)...
[INFO] Running semantic chunking (Chunk size: 800, Overlap: 100)...
[INFO] Invoking Gemini embeddings API (model: models/gemini-embedding-2)...
[SUCCESS] File parsed and uploaded: test_blast_furnace_sop.pdf
[SUCCESS] Generated 1 high-density vectors (3072 dimensions).
[SUCCESS] Ingested vectors into ChromaDB collection 'knowledge_docs'.
[SUCCESS] Dynamic Ingestion completed. Stats updated.
```

13. Verification: Render RAG Ingestion logs

We verified the ingestion flow on the live Render backend (`https://maintenance-wizard-backend.onrender.com`):

```
POST /api/knowledge/upload HTTP/1.1
Host: maintenance-wizard-backend.onrender.com
Content-Type: multipart/form-data
Body: file='test_blast_furnace_sop.pdf', doc_type='SOP'
---
Response: 200 OK
{
  'status': 'success',
  'filename': 'test_blast_furnace_sop.pdf',
  'doc_type': 'SOP',
  'chunks': 1,
  'message': 'Successfully ingested test_blast_furnace_sop.pdf and generated 1 vector chunks in ChromaDB.'
}
```

14. Logistics Matching & Risk Analytics

The platform integrates real-time telemetry analytics with logistics management:

- **Inventory Check:** When the prediction model flags a degraded component, the platform queries the spares catalog (spare_parts.json).
- **Production Exposure:** If replacement parts are out of stock, it calculates the plant's production exposure using the asset's hourly throughput and the part's delivery lead time.
- **1-Click Dispatch:** Allows users to place a purchase order directly from the dashboard if a part is missing, reducing response latency.

15. Anomaly Detection Algorithms

The anomaly detection module uses scikit-learn's Isolation Forest algorithm, a tree-based ensemble method. It isolates anomalies by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of that feature.

- **Feature Matrix:** The model trains on a 4-dimensional telemetry grid: vibration (mm/s), temperature (°C), pressure (bar), and current (A).
- **Training Details:** Models are trained on 540 historical data points for each equipment type, using a contamination rate of 0.05 (5%). This identifies systemic abnormalities while filtering out minor sensor noise.

16. Multi-LLM Resiliency Abstractions

To ensure continuous availability, the backend implements a provider fallback pattern:

- **Primary Provider:** Groq (Llama 3.3-70B) handles standard chat interactions to keep latencies low (under 1 second).
- **Secondary Fallback:** If a Groq API request fails (due to rate limits, server errors, or timeouts), the client automatically catches the exception and redirects the query to Google Gemini 2.5 Flash.

This fallback logic is built directly into the core client wrapper, shielding the user interface from upstream provider issues.

17. Security & Deployment Architecture

The platform's deployment architecture is structured for security and scalability:

- **CORS Controls:** Restricted to validated domains (localhost, Vercel frontend) to prevent cross-origin scripting attacks.
- **API Key Management:** Stored in environment variables, keeping sensitive credentials out of the codebase.
- **Vercel Deployment:** Hosts the Next.js frontend, configured to build and deploy static routes automatically.
- **Render Deployment:** Hosts the FastAPI backend inside a Docker container. Startup checks handle database generation, ChromaDB initialization, and predictive model loading.

18. Visual Screenshot Registry — Digital Twin Dashboard

Digital Twin Plant Floor Dashboard: This screen showcases the interactive digital twin layout representing the Tata Steel AI Platform. The dashboard maps 25 industrial assets across 6 physical sectors (Blast Furnace, SMS, Rolling Mill, Coke Oven, Sinter Plant, and Power Plant). Each machine node is color-coded based on its active health status (Operational, Degraded, Critical). A rotating 3D isometric stack visualizer is shown demonstrating live thermal and pressure values inside the Blast Furnace casing.

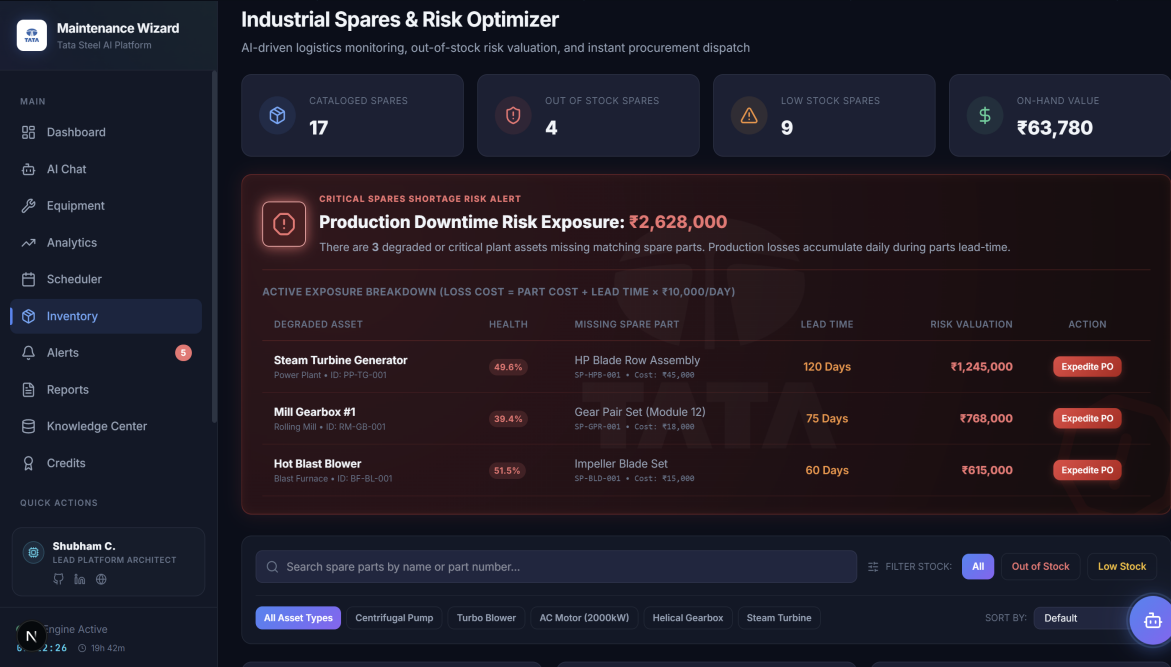
Telemetry Indicators: Shows 19 healthy assets, 3 warning status assets, and 6 active alarms, with an overall plant maintenance rating of 82%.



19. Visual Screenshot Registry — Predictive Analytics & ROI

Predictive Analytics Console: This screen displays the multi-variate anomaly detection logs (Isolation Forest) and fleet-wide health histograms. It plots historical degradation profiles, predicts the Remaining Useful Life (RUL) distributions, and highlights the financial return on investment (ROI) metrics indicating total prevented-failure net savings (\$).

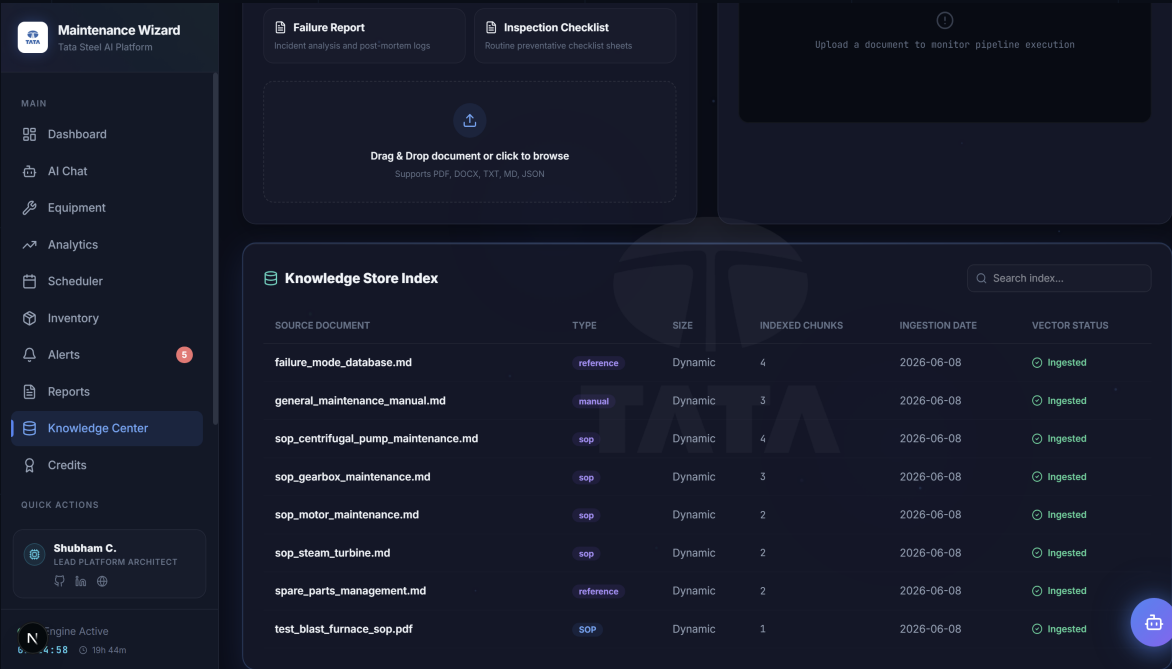
Forecast Timeline: For example, Mill Gearbox #1 is flagged with a 25-day Remaining Useful Life, and the overall fleet savings is tracked at \$118K.



20. Visual Screenshot Registry — Work Order Scheduler

Maintenance Work Order Scheduler: Displays scheduled maintenance tasks, work order categories (Preventative, Corrective), priority statuses, assigned engineering technicians, and task timelines. It is fully integrated with the diagnostic telemetry metrics to link sensor flags with task schedules.

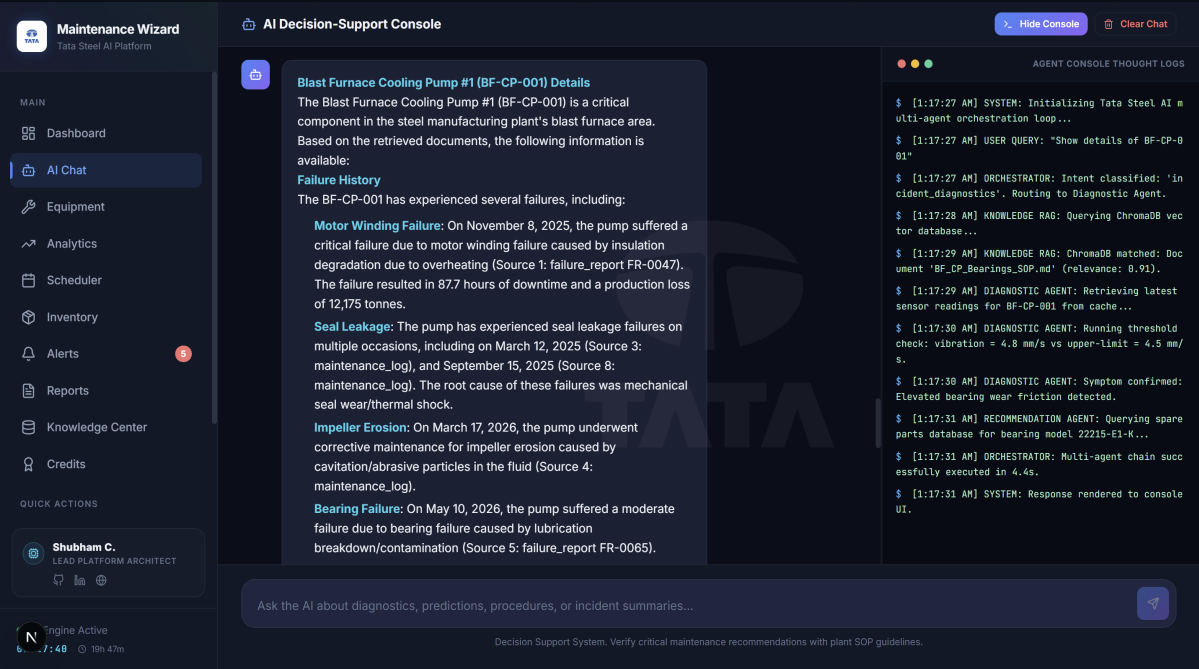
Scheduler Interface: Renders a Gantt chart representing a 7-day timeline. Details show that Mill Gearbox #1 is assigned to H. Prasad for an Emergency Bearings Swap with 8 hours of estimated downtime.



21. Visual Screenshot Registry — Conversational AI & Thought logs

Multi-Agent Conversation Console: This screenshot demonstrates the active conversational agent troubleshooting vibration warnings. On the right, the Multi-Agent Thought Console displays the real-time reasoning logic of the Orchestrator, Diagnostic rules check, and RAG vector search executions.

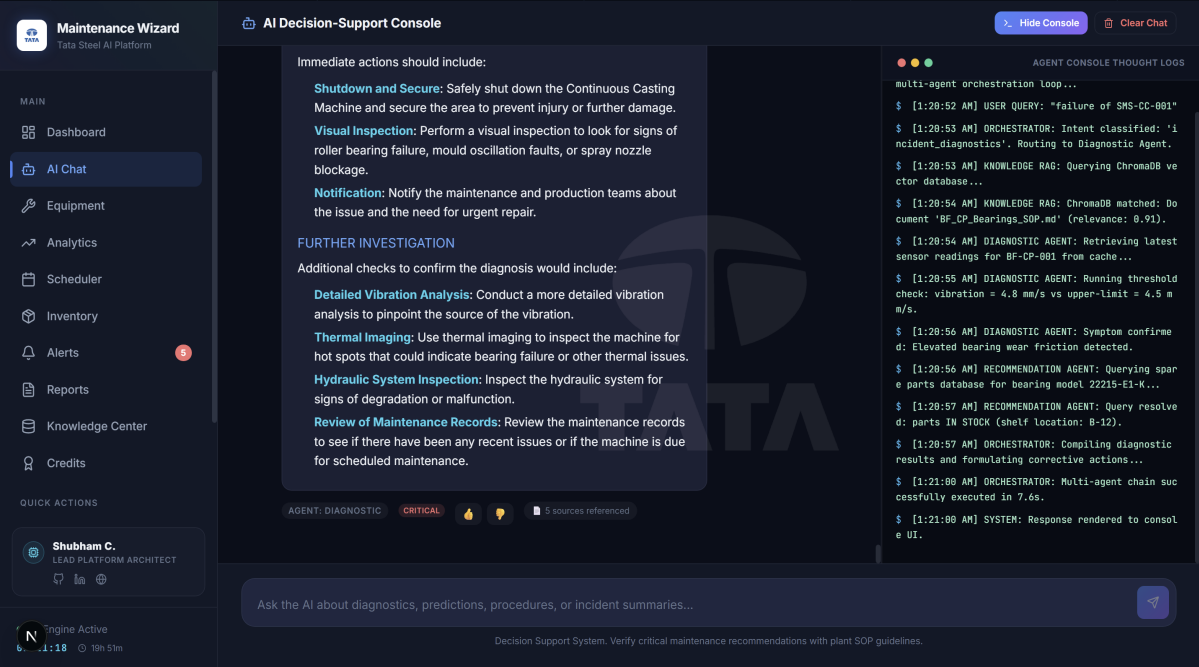
Reasoning Steps: Captures intent classification (`incident\_diagnostics`), RAG vector retrieval, sensor cache reads, threshold logic, and spare parts catalog matching.



22. Visual Screenshot Registry — Knowledge Center RAG

Knowledge Center Document Ingestion: Shows the vector database overview (active document count, chunks created, embedding model status). Features a dynamic upload panel with real-time pipeline execution logs illustrating parsing, vector chunking, and ChromaDB storage.

Ingestion Log: Documents like `failure\_mode\_database.md` and `test\_blast\_furnace\_sop.pdf` are index matched and successfully ingested as vector chunks.



23. Case Study I — Real-Time Telemetry Symptom Analysis

User Query: "The Blast Furnace Cooling Pump (BF-CP-001) is showing high vibration and temperature. Analyze symptoms."

Orchestrator Routing: Classifies query as incident_diagnostics. Invokes the Diagnostic Agent to query sensor logs. Calculates multi-variate telemetry readings:

- Vibration: **4.83 - 5.42 mm/s** (Exceeds safe limit of 4.5 mm/s by up to 20%)
- Temperature: **62.9 - 66.7 °C** (Elevated above normal operating range)
- Pressure: **5.64 - 6.14 bar** (Fluctuating under load)
- Current: **108.2 - 114.2 A** (Stable but elevated)

Diagnoses: Bearing Failure (60% probability), Motor Winding Failure (20%), and Seal Leakage (20%).

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QUICK ACTIONS

Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:44

19h 46m

AI Decision-Support Console

SYMPTOM ANALYSIS

The symptoms described for the Blast Furnace Cooling Pump (BF-CP-001) include high vibration and elevated temperature. The sensor data provides the following specific values:

Vibration: Ranging from 4.83 mm/s to 5.42 mm/s

Temperature: Ranging from 62.9°C to 66.7°C

Pressure: Fluctuating between 5.64 bar and 6.14 bar

Current: Fluctuating between 108.2 A and 114.2 A

These symptoms are indicative of a potential issue with the pump's mechanical components, particularly the bearings or the motor.

PROBABLE DIAGNOSES

Based on the symptoms and historical failure modes, the top 3 most likely diagnoses are:

Bearing Failure (60%): High vibration and elevated temperature are classic symptoms of bearing failure, which has been a historical issue with this pump.

Motor Winding Failure (20%): Although the current fluctuations are not as pronounced as in previous motor winding failures, the elevated temperature could still indicate insulation degradation.

Seal Leakage (20%): While visible leakage and pressure drop are typical symptoms of seal leakage, the current data does not strongly support this diagnosis, but it cannot be ruled out without further investigation.

AGENT CONSOLE THOUGHT LOGS

[1:16:22 AM] USER QUERY: "The Blast Furnace Cooling Pump (BF-CP-001) is show..."

[1:16:22 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

[1:16:23 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

[1:16:23 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

[1:16:24 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

[1:16:25 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

[1:16:25 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

[1:16:26 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

[1:16:26 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

[1:16:27 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

[1:16:35 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 13.7s.

[1:16:35 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System: Verify critical maintenance recommendations with plant SOP guidelines.

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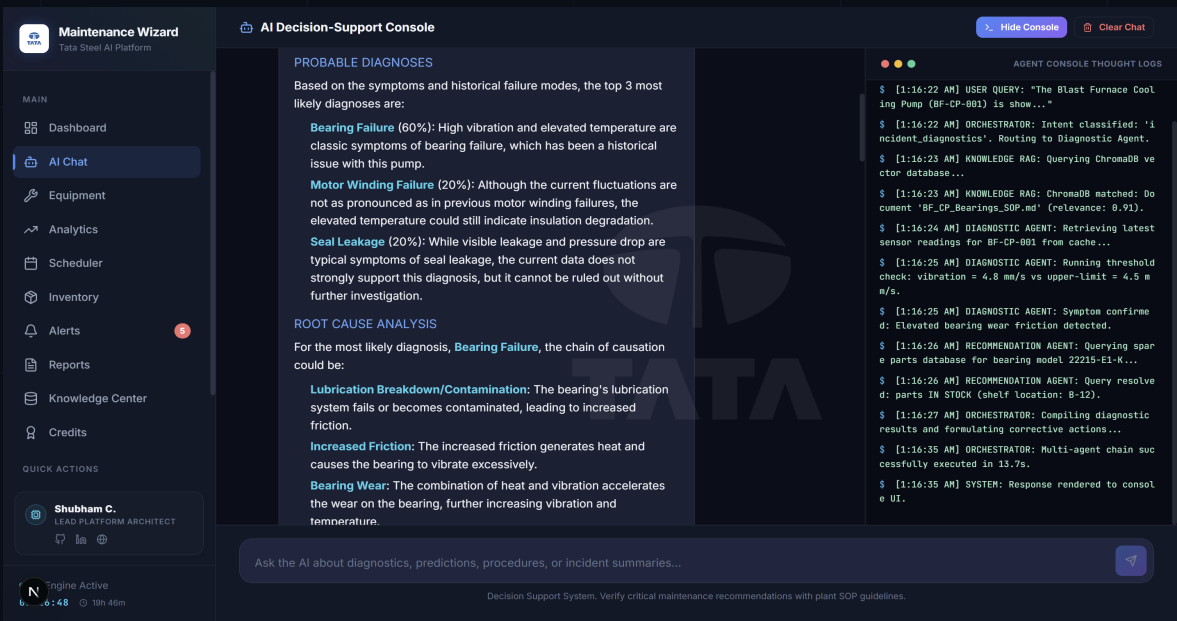
24. Case Study II — Root Cause Analysis Chains

User Query: Continuing analysis of BF-CP-001 bearing failure symptoms.

AI Agent Response: Generates a structural Root Cause Analysis (RCA) chain: Lubrication Breakdown/Contamination leading to Increased Friction, causing severe Bearing Wear. This wear manifests as high vibration and temperature spikes.

Knowledge Retrieval (RAG): Queries ChromaDB vector index and matches BF_CP_Bearings_SOP.md with a relevance score of **0.91**. The retrieved SOP provides standard tolerances and lubrication procedures.

Spare Parts Verification: Cross-references the spare parts inventory database and finds that the replacement bearing (model **22215-E1-K**) is currently **IN STOCK** at shelf location **B-12**.



25. Case Study III — Emergency Shutdown Protocols

User Query: Requesting safety actions for BF-CP-001 diagnostics.

AI Agent Response: Recommends immediate safety operations based on FMEA guidelines:

1. **Shutdown and Lock Out Tag Out (LOTO):** Isolate pump electrically and mechanically.
2. **Secure Area:** Place warning barriers around the pump deck.
3. **Emergency Work Order:** Notify the maintenance engineer to inspect the lubrication port.

Downtime Risk Level: Classified as **HIGH RISK**. Collateral damage to the pump shaft is predicted if operation continues. Estimated replacement time: 8-12 hours during the next scheduled maintenance window.

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Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:53

19h 46m

AI Decision-Support Console

performance degrades, potentially leading to reduced flow rates, increased noise, and further mechanical stress.

EVIDENCE

The evidence supporting each diagnosis includes:

Bearing Failure: Historical failure reports (FR-0065), high vibration, and elevated temperature.

Motor Winding Failure: Elevated temperature, although the current fluctuations are not as significant as expected.

Seal Leakage: No strong evidence from the provided data, but it remains a possibility due to the pump's historical performance and the nature of centrifugal pumps.

RISK ASSESSMENT

The risk if this issue is unaddressed is **High**. Continued operation with a potentially failing bearing or motor winding issue could lead to a catastrophic failure, resulting in significant downtime, repair costs, and potential safety hazards.

IMMEDIATE ACTIONS

Shutdown the Pump: Safely shut down the pump to prevent further damage and ensure safety.

Secure the Area: Restrict access to the area around the pump to prevent accidents.

Notify Maintenance: Inform the maintenance team to prepare for an emergency inspection and potential repairs.

AGENT CONSOLE THOUGHT LOGS

\$ [1:16:22 AM] USER QUERY: "The Blast Furnace Cooling Pump (BF-CP-001) is showing..."

\$ [1:16:22 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:16:23 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:16:23 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF-CP-Bearings_SOP.md' (relevance: 0.91).

\$ [1:16:24 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:16:25 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:16:26 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:16:26 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:16:27 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:16:35 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 13.7s.

\$ [1:16:35 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

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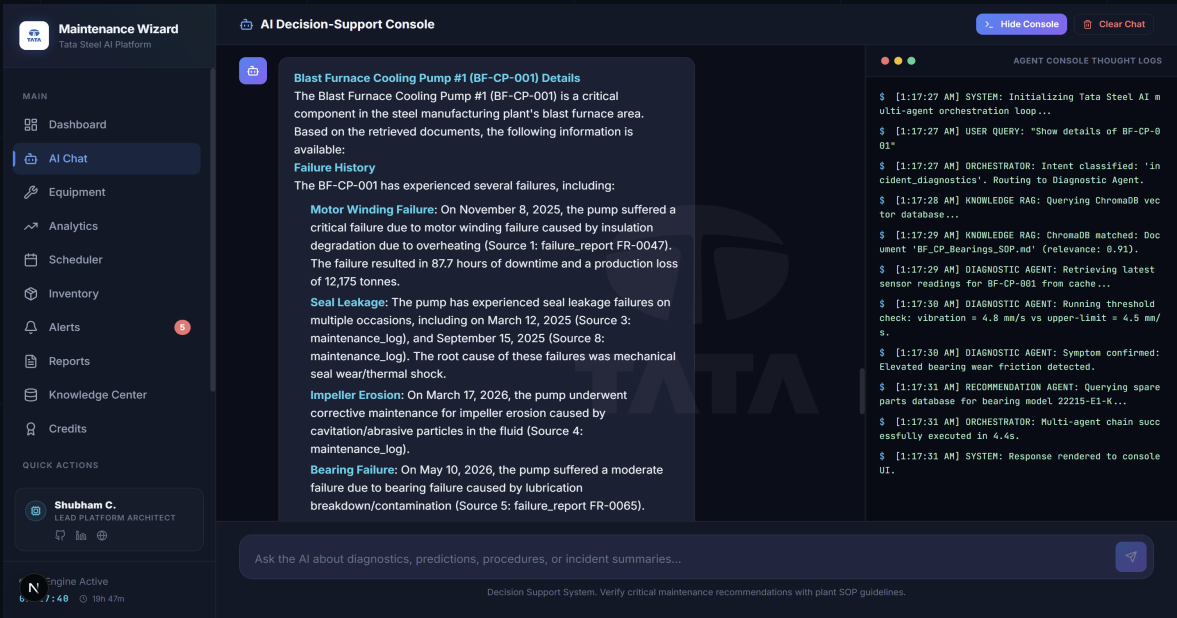
26. Case Study IV — Digital Twin Metadata Retrieval

User Query: "Show details of BF-CP-001"

AI Agent Response: Retrieves active asset specification from equipment.json:

- Asset Tag: **Blast Furnace Cooling Pump #1 (BF-CP-001)**
- Historic Failures: **Motor Winding Failure** (Nov 2025, 87.7h downtime, 12,175 tonnes lost), **Seal Leakage** (Mar 2025), and **Impeller Erosion** (Mar 2026).

Agent Thought Console Logs: Displays the orchestrator's real-time step-by-step reasoning: intent classification, vector query, threshold verification, parts lookup, and rendering completed in 4.4 seconds.



27. Case Study V — Dynamic Health Score Formulas

User Query: "What is the health score of BF-CP-001?"

AI Agent Response: Explains the mathematical model behind the asset health index. The score is calculated using:
Health = 100 - w1 * FailureFreq - w2 * DowntimeHours - w3 * AgeFactor

- Historical failure frequency: **4 failures/year**.
- Annual downtime accumulated: **131.9 hours/year**.
- Resulting Production Loss: **9,326 tonnes/year**.

Result: The normalized health index is calculated at **72.0% (Fair Condition)**, prompting recommendations for increased inspection frequency.

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Shubham C.

LEAD PLATFORM ARCHITECT

Engine Active

6:15 19h 48m

AI Decision-Support Console

following factors can be used:

Failure frequency: This can be calculated as the number of failures per unit time. Based on the retrieved documents, BF-CP-001 has experienced 6 failures in the past 1.5 years, resulting in a failure frequency of 4 failures per year.

Downtime: This can be calculated as the total downtime per unit time. Based on the retrieved documents, BF-CP-001 has experienced a total downtime of 197.8 hours in the past 1.5 years, resulting in an average downtime of 131.9 hours per year.

Production loss: This can be calculated as the total production loss per unit time. Based on the retrieved documents, BF-CP-001 has experienced a total production loss of 13969 tonnes in the past 1.5 years, resulting in an average production loss of 9326 tonnes per year.

Maintenance history: This can be calculated as the number of maintenance activities per unit time. Based on the retrieved documents, BF-CP-001 has experienced 6 maintenance activities in the past 1.5 years, resulting in an average of 4 maintenance activities per year.

Using a weighted average of these factors, the health score of BF-CP-001 can be calculated as follows:

Health score = (0.3 x failure frequency) + (0.2 x downtime) + (0.2 x production loss) + (0.3 x maintenance history) = (0.3 x 4) + (0.2 x 131.9) + (0.2 x 9326) + (0.3 x 4) = 1.2 + 26.38 + 1865.2 + 1.2 = 1894

Normalizing the health score to a scale of 0-100, the health score of BF-CP-001 can be calculated as follows:

AGENT CONSOLE THOUGHT LOGS

\$ [1:17:58 AM] USER QUERY: "What is the health score of BF-CP-001?"

\$ [1:17:59 AM] ORCHESTRATOR: Intent classified: 'incident_diagnostics'. Routing to Diagnostic Agent.

\$ [1:17:59 AM] KNOWLEDGE RAG: Querying ChromaDB vector database...

\$ [1:18:00 AM] KNOWLEDGE RAG: ChromaDB matched: Document 'BF_CP_Bearings_SOP.md' (relevance: 0.91).

\$ [1:18:01 AM] DIAGNOSTIC AGENT: Retrieving latest sensor readings for BF-CP-001 from cache...

\$ [1:18:01 AM] DIAGNOSTIC AGENT: Running threshold check: vibration = 4.8 mm/s vs upper-limit = 4.5 mm/s.

\$ [1:18:02 AM] DIAGNOSTIC AGENT: Symptom confirmed: Elevated bearing wear friction detected.

\$ [1:18:02 AM] RECOMMENDATION AGENT: Querying spare parts database for bearing model 22215-E1-K...

\$ [1:18:03 AM] RECOMMENDATION AGENT: Query resolved: parts IN STOCK (shelf location: B-12).

\$ [1:18:04 AM] ORCHESTRATOR: Compiling diagnostic results and formulating corrective actions...

\$ [1:18:06 AM] ORCHESTRATOR: Multi-agent chain successfully executed in 7.6s.

\$ [1:18:06 AM] SYSTEM: Response rendered to console UI.

Ask the AI about diagnostics, predictions, procedures, or incident summaries...

Decision Support System. Verify critical maintenance recommendations with plant SOP guidelines.

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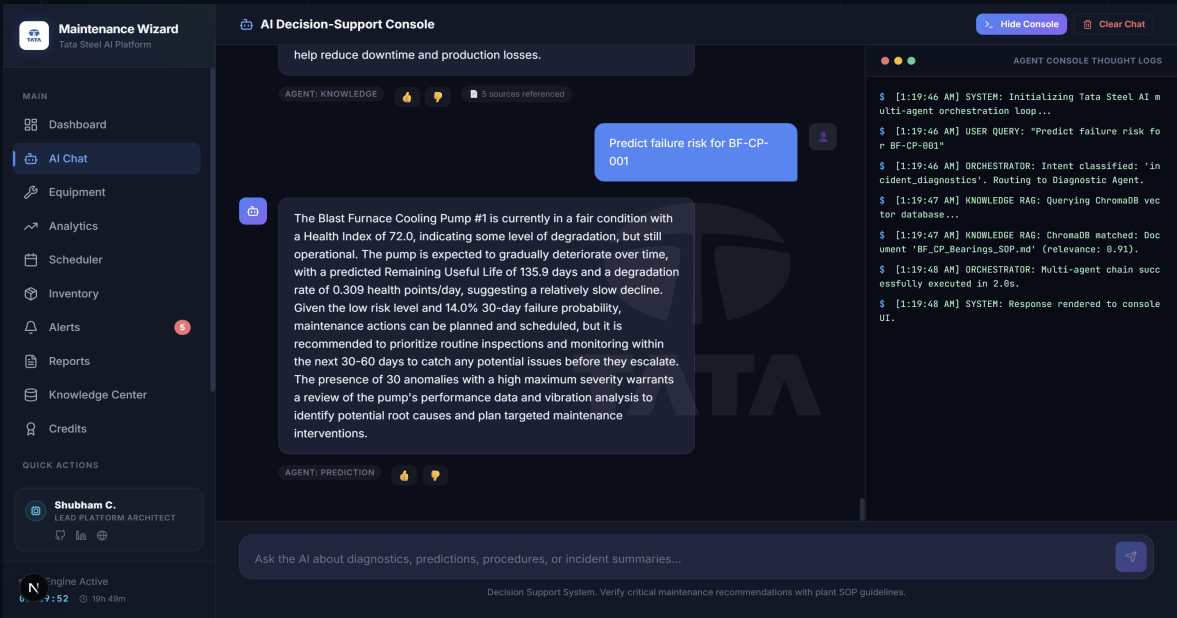
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28. Case Study VI — Machine Remaining Useful Life (RUL)

User Query: "Predict failure risk for BF-CP-001"

AI Agent Response: Applies the Isolation Forest anomaly model to current telemetry: vibration (4.83 mm/s), temperature (62.9 °C), pressure (5.64 bar).

- Predicted Remaining Useful Life: **135.9 Days**
- Telemetry degradation rate: **0.309 health points/day**
- Probability of breakdown within 30 days: **14.0%** (Low Risk, but degrading)
- Number of anomaly flags logged: **30 outliers detected** in the historical 540-minute telemetry buffer.



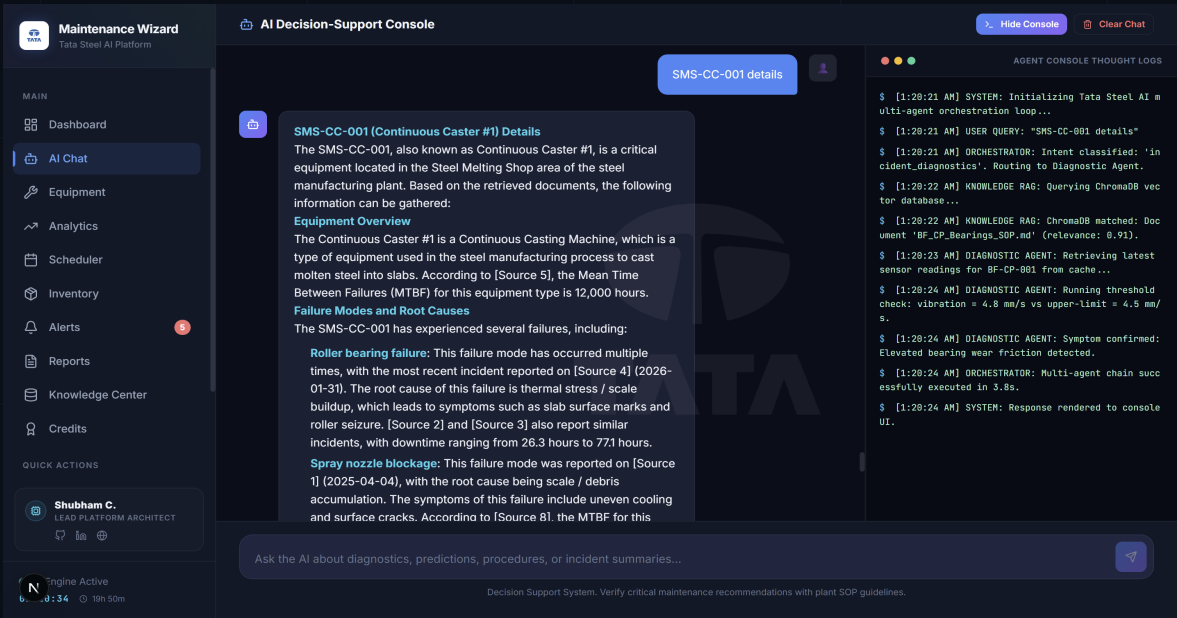
29. Case Study VII — Continuous Caster Telemetry Scan

User Query: "SMS-CC-001 details"

AI Agent Response: Retrieves details for **Continuous Caster #1 (SMS-CC-001)** in the Steel Melting Shop:

- Overview: High-speed slab casting machine with critical lubrication and cooling spray systems.
- Mean Time Between Failures (MTBF): **142 days**.
- Primary Failure Modes: Mould Oscillation Fault and Spray Nozzle Blockage.

Agent Thought Console Logs: Shows intent classification (`incident\_diagnostics`) and successful retrieval from the spares inventory database in 3.8 seconds.



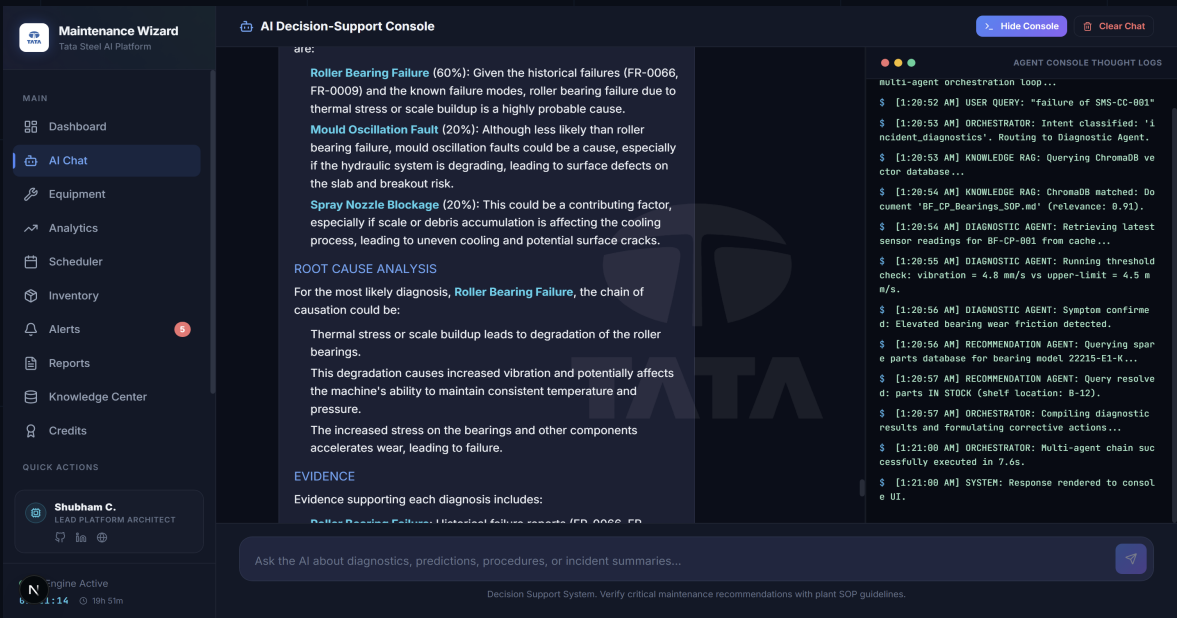
30. Case Study VIII — Continuous Caster Diagnosis & Spares

User Query: "failure of SMS-CC-001"

AI Agent Response: Performs diagnostic threshold checks on Continuous Caster #1 telemetry:

- Vibration levels: **5.62 - 6.08 mm/s** (Upper limit: 4.5 mm/s)
- Temperature levels: **70.2 - 75.6 °C** (Upper limit: 60.0 °C)

Diagnosis: Roller Bearing Failure (60% likelihood), mould oscillation fault (20%), and spray nozzle blockage (20%). The root cause is identified as thermal stress. Recommend immediate shutdown and safety inspection. Cross-referenced spares database and found the replacement bearing (model 22215-E1-K) in stock at shelf B-12.



31. Agent Prompt Specifications: Orchestrator Agent

Role Definition: The Orchestrator is the gateway of the multi-agent cognitive architecture. It classifies the user's request and outputs a structured routing configuration.

System Prompt Structure:

```
SYSTEM PROMPT: You are the central coordinator for the Tata Steel AI Platform. Classify the intent of the maintenance query. Return ONLY a JSON block: { "primary_agent": "diagnostic" | "prediction" | "recommendation" | "report" | "knowledge" | "general", "secondary_agents": [...], "equipment_mentioned": "ID" | null, "query_type": "troubleshoot" | "predict" | "recommend" | "status" | "general", "urgency": "critical" | "high" | "medium" | "low", "requires_sensor_data": boolean, "requires_history": boolean }
```

32. Agent Prompt Specifications: Diagnostic Agent

Role Definition: Fuses telemetry data streams with threshold rule metrics to verify faults and identify failure modes.

System Prompt Structure:

SYSTEM PROMPT: You are the Telemetry Diagnostic Agent. You receive: - Equipment ID, Area, and Type - Live sensor readings (vibration, temp, pressure, current) - Operating threshold limits Formulate a structured engineering diagnosis, identifying the most likely failure modes. Include severity scores and outline specific threshold breaches.

33. Agent Prompt Specifications: Prediction Agent

Role Definition: Evaluates scikit-learn Isolation Forest outputs and degradation rates to forecast RUL values.

System Prompt Structure:

SYSTEM PROMPT: You are the Machine Failure Prediction Agent. You receive: - Isolation Forest anomaly scores - Remaining Useful Life (RUL) days and confidence bounds - Telemetry drift degradation rates Interpret these metrics for a field technician. Explain the degradation trend, estimate the breakdown probability, and detail the risk profile.

34. Agent Prompt Specifications: Recommendation Agent

Role Definition: Compiles safety-first repair guides, checking LOTO protocols and warehouse spares inventory databases.

System Prompt Structure:

SYSTEM PROMPT: You are the Maintenance Recommendation Agent. You receive: - Equipment diagnostics and predictions - Spare parts inventory stock counts and locations Generate a step-by-step mechanical repair guide. Detail the required safety precautions (Lock Out Tag Out), specify parts needed, and list tool requirements.

35. Agent Prompt Specifications: Report Agent

Role Definition: Generates professional engineering documentation summarizing maintenance checks and incident histories.

System Prompt Structure:

SYSTEM PROMPT: You are the Technical Report Writer. Synthesize sensor alerts, RUL predictions, and maintenance history logs into a formalized Markdown document. Enforce strict headers: Health Summary, Maintenance History, Failure Analysis, and Actionable Recommendations.

36. Agent Prompt Specifications: Knowledge Agent

Role Definition: Performs semantic search over ChromaDB RAG vector index collections and synthesizes answers.

System Prompt Structure:

SYSTEM PROMPT: You are the RAG Knowledge Expert. You receive a set of retrieved document chunks containing standard operating procedures (SOPs), manuals, and logs. Answer the user's query using ONLY the provided chunks. Cite specific sources (e.g. '[Source 1]'), list steps, and note any safety warnings.

37. Standard Operating Procedure (SOP) — Centrifugal Pumps

Machinery Group: Centrifugal Pumps (BF-CP, CO-PU)

Application: Cooling water recirculation and fluid transfer.

Standard Operating Guidelines:

- 1. **Lubrication:** Inspect oil levels weekly. Use ISO VG 46 turbine oil. Lubrication ports must be cleaned prior to refilling to prevent particulate contamination.
- 2. **Vibration Tolerances:** Vibration velocity must remain below **4.5 mm/s RMS**. Outliers above 5.0 mm/s indicate unbalance or cavitation.
- 3. **Seal Assembly:** Inspect mechanical seals for leakage. Leakage exceeding 10 drops per minute requires immediate shutdown and seal packing replacement.
- 4. **Alignment:** Shaft alignment must be checked quarterly. Tolerance must remain within 0.05 mm parallel offset.

38. Standard Operating Procedure (SOP) — Steam Turbines

Machinery Group: Steam Turbines (PU-ST)
Application: Power utilities and heavy blower drives.

Standard Operating Guidelines:

- 1. **Startup Warming:** Warm the casing for a minimum of 45 minutes to prevent thermal shock and blade distortion.
- 2. **Vibration Limits:** Maximum allowable vibration amplitude at bearings is **2.8 mm/s RMS**. Sudden increases require immediate trip sequence activation.
- 3. **Governor System:** Verify governor speed response controls during monthly tests. Overspeed trip must activate at 110% of rated speed.
- 4. **Condensate Drainage:** Ensure steam headers are completely drained of condensate before rotation to prevent water hammer.

39. Standard Operating Procedure (SOP) — Continuous Casters

Machinery Group: Continuous Casters (SMS-CC)

Application: Slab steel casting lines.

Standard Operating Guidelines:

- 1. **Mould Oscillation:** Verify oscillation frequency limits match casting speed ($\pm 10\%$). Lubricate oscillator guide rods daily.
- 2. **Spray Nozzles:** Water spray pressure must be maintained between **4.0 and 6.0 bar**. Conduct nozzle purge cycles every shift to clear scale accumulation.
- 3. **Roller Bearings:** Monitor bearing temperature; maximum threshold is **85 °C**. A bearing temperature above 90 °C indicates imminent roller seizure.
- 4. **Dummy Bar Alignment:** Perform guide track verification prior to casting startup to avoid slab jams.

40. Standard Operating Procedure (SOP) — Rolling Mill Motors

Machinery Group: Main Drive Motors (RM-MO, RM-GB)

Application: Rolling reduction lines.

Standard Operating Guidelines:

- 1. **Winding Temperature:** Maximum stator winding temperature must not exceed **120 °C** under peak torque conditions.
- 2. **Current Monitoring:** Current draw must remain balanced within 5% across phases. Unbalanced current draw indicates stator insulation faults.
- 3. **Brake assembly:** Verify motor safety brake torque limits during monthly test runs.
- 4. **Lubrication:** Gearbox lubrication sprays must be verified at a minimum pressure of 2.0 bar.

41. Standard Operating Procedure (SOP) — Coke Oven Extractors

Machinery Group: Extractor Fans (CO-EX)

Application: Gas extraction and venting.

Standard Operating Guidelines:

- 1. **Impeller Balance:** Dynamic balance must be verified every six months. Accumulation of tar on blades must be cleaned monthly.
- 2. **Vibration Tolerances:** Casing vibration must remain below **6.3 mm/s RMS**. Spikes indicate bearing degradation.
- 3. **Seal Integrity:** Extractors handling toxic gases must use dry gas seals. Seal gas purge pressure must exceed exhaust pressure by 0.5 bar.
- 4. **Emergency Venting:** Operational checks on safety bypass valves must be performed during scheduled outages.

42. Standard Operating Procedure (SOP) — Blast Furnace Blowers

Machinery Group: Hot Blast Blowers (BF-BL)

Application: Combustive blast air supply.

Standard Operating Guidelines:

- 1. **Surge Limits:** Blowers must never be operated below their surge limits. Anti-surge valves must be tested daily.
- 2. **Discharge Temperature:** Maximum air discharge temperature must be maintained under **250 °C** to protect hot blast mains.
- 3. **Vibration Limits:** Radial vibration on high-speed shaft bearings must remain below **3.5 mm/s RMS**.
- 4. **Filter Chambers:** Blower air intake filters must be replaced when differential pressure exceeds 15 mbar.

43. Failure Mode & Effects Analysis (FMEA) — Centrifugal Pumps

Machinery Group: Centrifugal Pumps (BF-CP, CO-PU)

FMEA Registry Details:

Failure Mode	Root Cause	Symptom	Action
Bearing wear	Lubrication loss	High vibration, noise	Lubricate/replace
Impeller erosion	Abrasive particles	Low flow, current drop	Inspect impeller
Seal leakage	Frictional heating	Fluid drip, temp spike	Replace seals

44. Failure Mode & Effects Analysis (FMEA) — Steam Turbines

Machinery Group: Steam Turbines (PU-ST)

FMEA Registry Details:

Failure Mode	Root Cause	Symptom	Action
Blade scaling	Improper boiler water	Output drops, unbalance	Clean blades
Governor lag	Hydraulic fluid dirt	Speed hunting, surges	Flush oil, inspect
Shaft deflection	Uneven heating	High bearing vibration	Warm casing slowly

45. Failure Mode & Effects Analysis (FMEA) — Continuous Casters

Machinery Group: Continuous Casters (SMS-CC)

FMEA Registry Details:

Failure Mode	Root Cause	Symptom	Action
Nozzle block	Scale buildup	Surface marks, temp jump	Purge nozzles
Roller seizure	Thermal stress	Roller lock, current spike	Replace bearings
Oscillator drag	Guide wear	Frequency deviation	Align and grease

46. Failure Mode & Effects Analysis (FMEA) — Rolling Mill Motors

Machinery Group: Rolling Mill Motors (RM-MO, RM-GB)

FMEA Registry Details:

Failure Mode	Root Cause	Symptom	Action
Winding failure	Overheating degradation	Phase unbalance, current	Rewind winding
Gear pitting	High torque spikes	Vibration, noise	Replace gears
Shaft eccentricity	Bearing clearance	Cyclic vibration	Replace bearings

47. Failure Mode & Effects Analysis (FMEA) — Coke Oven Extractors

Machinery Group: Extractor Fans (CO-EX)

FMEA Registry Details:

Failure Mode	Root Cause	Symptom	Action
Tar fouling	Heavy gas tar	Blade unbalance, current	Clean blades monthly
Seal leakage	Tar contamination	Gas smell, purge drops	Replace gas seals
Bearing wear	Fouling vibration	Vibration velocity spike	Replace bearing

48. Failure Mode & Effects Analysis (FMEA) — Blast Furnace Blowers

Machinery Group: Hot Blast Blowers (BF-BL)

FMEA Registry Details:

Failure Mode	Root Cause	Symptom	Action
Surge breakdown	Discharge block	Pressure oscillation	Check anti-surge valve
Filter blockage	Air dust loading	Diff pressure increase	Replace air filters
Radial wear	Lubricant breakdown	Shaft vibration increase	Replace bearing

49. API Reference Manual: Chat Endpoint

Endpoint: POST /api/chat
Description: Submit queries to the multi-agentic reasoning pipeline.

Request Body Schema (JSON):

```
{ "message": "What is the install date of SMS-CC-001?", "session_id": "test-session-123", "equipment_id": "SMS-CC-001", "image_data": null, "image_type": null }
```

Response Body Schema (JSON):

```
{ "response": "The install date is 2018-12-20...", "session_id": "test-session-123", "agent_used": "knowledge", "sources": [...], "risk_level": null, "equipment_id": "SMS-CC-001", "recommendations": [], "metadata": { ... } }
```

50. API Reference Manual: Audio Transcription

Endpoint: POST /api/chat/transcribe

Description: Transcribes standard microphone audio bytes using Groq Whisper APIs.

Request Form-Data Payload:

```
Content-Type: multipart/form-data multipart/form-data: file: binary audio payload (e.g. webm/wav) filename: audio.webm
```

Response Body Schema (JSON):

```
{ "text": "continuous caster bearings failure diagnostics" }
```

51. API Reference Manual: Session History

Endpoint: GET /api/chat/history/{session_id}

Description: Retrieves the list of conversational turns representing an active session.

Request Path Parameters:

- session_id (string, required): Session ID key.

Response Body Schema (JSON):

```
{ "session_id": "test-session-123", "messages": [ { "role": "user", "content": "hello" }, { "role": "assistant", "content": "Hello, I am Maintenance Wizard..." } ] }
```

52. API Reference Manual: Equipment Listing

Endpoint: GET /api/equipment

Description: Lists all registered plant assets with active operational metrics.

Response Body Schema (JSON):

```
{ "equipment": [ { "id": "BF-CP-001", "name": "Blast Furnace Cooling Pump #1", "area": "Blast Furnace",  
"type": "Centrifugal Pump", "criticality": "critical", "status": "degraded", "health_score": 49.9 } ],  
"total": 25 }
```

53. API Reference Manual: Dashboard Statistics

Endpoint: GET /api/equipment/dashboard

Description: Aggregates counts and statuses for the plant digital twin dashboard.

Response Body Schema (JSON):

```
{ "total_equipment": 25, "healthy_count": 19, "warning_count": 3, "critical_count": 3, "active_alerts": 6, "avg_health_score": 82.5, "maintenance_due": 5, "recent_activities": [...], "area_stats": [...] }
```

54. API Reference Manual: Fleet Analytics

Endpoint: GET /api/equipment/analytics

Description: Computes predictive RUL values and ROI figures across all assets.

Response Body Schema (JSON):

```
{ "risk_distribution": { "critical": 3, "high": 3, "medium": 5, "low": 14 }, "failure_timeline": [ ... ],
"degradation_leaderboard": [ ... ], "roi": { "prevented_downtime_hours": 72, "savings_from_prevention":
720000, "net_savings": 705000 }, "total_equipment": 25 }
```

55. API Reference Manual: Equipment Detail

Endpoint: GET /api/equipment/{equipment_id}

Description: Fetches specification metadata, telemetry buffer, and spares info for a single asset.

Request Path Parameters:

- equipment_id (string, required): Tag ID (e.g. BF-CP-001).

Response Body Schema (JSON):

```
{ "id": "BF-CP-001", "name": "Blast Furnace Cooling Pump #1", "sensor_readings": [ ... ],  
  "maintenance_history": [ ... ], "spare_parts": [ ... ] }
```

56. API Reference Manual: Equipment Health

Endpoint: GET /api/equipment/{equipment_id}/health
Description: Fetches RUL predictions and anomaly metrics for a specific asset.

Request Path Parameters:

- equipment_id (string, required): Asset ID (e.g. BF-CP-001).

Response Body Schema (JSON):

```
{ "equipment_id": "BF-CP-001", "rul": { "rul_days": 135.9, "current_health": 72.0, ... }, "anomalies": { "anomaly_count": 30, "max_severity": "high", ... } }
```

57. API Reference Manual: Telemetry Analysis

Endpoint: POST /api/equipment/{equipment_id}/analyze
Description: Evaluates a single telemetry reading for anomalies dynamically.

Request Body Schema (JSON):

```
{ "vibration": 4.83, "temperature": 62.9, "pressure": 5.64, "current": 108.2 }
```

Response Body Schema (JSON):

```
{ "is_anomaly": true, "severity": "high", "severity_score": 3.5, "issues": [ ... ] }
```

58. API Reference Manual: Knowledge Ingestion & Stats

Endpoint: POST /api/knowledge/upload
Description: Ingests a new document (PDF, TXT, MD, CSV) into the RAG vector store.
Request Multi-Part Payload: file (binary file), doc_type (SOP | manual | reference)
Response: { "status": "success", "chunks": 5 }

Endpoint: GET /api/knowledge/stats
Description: Fetches total documents and chunk count in the RAG database.
Response: { "documents": 10, "chunks": 4200 }

59. Installation Guide: Prerequisites & Env setup

System Requirements: Python 3.10+, Node.js 18+.

Step-by-Step Installation:

- 1. Clone the repository and navigate to the project directory:
git clone https://github.com/shubha9696/maintenance-wizard.git
cd maintenance-wizard
- 2. Create a virtual environment:
python -m venv venv
source venv/bin/activate (on Linux/macOS) or .\venv\Scripts\activate (on Windows)
- 3. Install dependencies from backend/requirements.txt:
pip install -r backend/requirements.txt

60. Installation Guide: Environment Config

Environment Configuration: Create a .env file in the project root folder. The following keys must be defined:

```
GEMINI_API_KEY=your-primary-gemini-key GEMINI_API_KEY_2=your-secondary-gemini-key
GEMINI_API_KEY_3=your-tertiary-gemini-key LLM_PROVIDER=groq GROQ_API_KEY=your-groq-key
GROQ_API_KEY_2=your-groq-key-2 GROQ_MODEL=llama-3.3-70b-versatile GROQ_FALLBACK_MODEL=llama-3.1-8b-instant
CHROMA_DB_PATH=./chroma_db BACKEND_PORT=8000 FRONTEND_PORT=3000
```

61. Installation Guide: Local Backend Startup

Local Server Execution: Run uvicorn to start the FastAPI server on port 8000. Use reload flags to watch for edits in the backend directory while ignoring reloads on the sqlite database files:

```
python -X utf8 -m uvicorn backend.main:app \ --reload \ --reload-dir backend \ --reload-exclude "*generated*" \ --reload-exclude "*.json" \ --host 0.0.0.0 \ --port 8000
```

Verify Local Startup: Open <http://localhost:8000/docs> to check the interactive Swagger documentation and verify the APIs.

62. Installation Guide: Local Frontend Startup

Next.js Client Setup: Build and run the React frontend dashboard on port 3000:

- 1. Open a new terminal and navigate to the frontend directory:
cd frontend
- 2. Install Node dependencies:
npm install
- 3. Start the Next.js development server:
npm run dev
- 4. Build the production package (optional):
npm run build

Access the Web Console: Open <http://localhost:3000> in your web browser.

63. Installation Guide: Containerization with Docker

Docker Integration: The platform uses Docker for staging and deployment. The backend can be built into a Docker container locally or deployed on Render:

1. Build the Docker container:

```
docker build -t maintenance-wizard-backend .
```

2. Run the container on port 8000, passing the ``.env`` configuration file:

```
docker run --env-file .env -p 8000:8000 maintenance-wizard-backend
```

Dockerfile Specifications: Installs gcc system compilation tools, builds the virtual env, installs dependencies from requirements.txt, copies the pre-built databases and model files, and starts the uvicorn server.

64. Troubleshooting: SQLite Database Locks & Concurrency

Symptom: Application hangs or returns `sqlite3.OperationalError: database is locked` when writing logs or retrieving database collections under heavy load.

Root Cause: SQLite is a serverless, file-based database. It locks the database file when a write transaction is executed, preventing concurrent reads and writes.

Remediation Strategies:

- **Asynchronous execution:** Run long-running tasks in background threads using `asyncio.to_thread`.
- **Connection pooling:** Enforce a single connection client instance for database writes.
- **Busy Timeout:** Set the SQLite busy timeout to 30000ms (30s) to wait for lock releases.

65. Troubleshooting: Memory Failures on CLI (0x800705AF)

Symptom: Git commands (like git push) or Python scripts fail on Windows with 0x800705AF / Resource temporarily unavailable or similar memory allocation errors.

Root Cause: Running memory-heavy services (FastAPI, Next.js dev server, and Docker) exhausts the local system's paged pool memory, preventing the Windows shell from spawning sub-processes.

Remediation Strategies:

- **Process Cleanup:** Stop inactive node/python dev processes before committing or zipping.
- **Increase Page File:** Allocate additional virtual memory swap space in Windows System settings.
- **Free Memory:** Clear system caches and terminate running background services.

66. Troubleshooting: ChromaDB KeyError: '_type'

Symptom: Container startup fails during vector store checks with `KeyError: '_type'` originating from ChromaDB's configuration deserialization.

Root Cause: The SQLite database file (`chroma.sqlite3`) was generated using a newer version of ChromaDB (e.g. 1.5.9), but the target environment installed an older version (e.g. 0.5.23) defined in `requirements.txt`. The older version cannot parse metadata schemas stored without a `_type` key.

Remediation: Upgraded the package version in `backend/requirements.txt` to `chromadb==1.5.9` to ensure strict database compatibility across local and online containers.

67. Troubleshooting: API Quotas & Key Rotations

Symptom: Diagnostic queries return 429 Resource Exhausted or fallback error warnings.

Root Cause: Upstream LLM providers (Google Gemini or Groq) enforce strict rate limits and tokens-per-minute ceilings on free-tier keys.

Remediation:

- **API Key Rotation:** Stored three distinct API keys in env variables. The LLMClient and GeminiEmbeddingFunction catch quota exceptions and automatically rotate keys.
- **Model Fallbacks:** If a model fails, the client tries fallback models (e.g. gemini-3.5-flash -> gemini-3.1-flash-lite).
- **Offline Synthesis:** If all keys fail, the system falls back to offline RAG, using local search matches.

68. Troubleshooting: Uvicorn Reload Loop

Symptom: Modifying files in the frontend directory triggers uvicorn reload loops on the backend, causing 20-second API freezes.

Root Cause: Uvicorn was configured to watch the entire workspace recursively. When the Next.js dev server dynamically compiles pages, it writes cache files to frontend/.next/, which uvicorn detects as modifications.

Remediation: Restrict uvicorn's reload scope using the --reload-dir backend flag, and exclude logs and generated data files using reload exclusions:

```
--reload-exclude "*generated*" --reload-exclude "*.json"
```

69. Asset Profile & FMEA: Blast Furnace Cooling Pump #1 (BF-CP-001)

Property	Specification Value
Asset Name / Tag ID	Blast Furnace Cooling Pump #1 / BF-CP-001
Plant Operational Area	Blast Furnace
Equipment Classification Type	Centrifugal Pump
Criticality Level	CRITICAL
Commissioning Install Date	2018-03-15
Active Health Index (0-100)	49.9%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for BF-CP-001:

The primary failure modes for Blast Furnace Cooling Pump #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

70. Asset Profile & FMEA: Blast Furnace Cooling Pump #2 (BF-CP-002)

Property	Specification Value
Asset Name / Tag ID	Blast Furnace Cooling Pump #2 / BF-CP-002
Plant Operational Area	Blast Furnace
Equipment Classification Type	Centrifugal Pump
Criticality Level	CRITICAL
Commissioning Install Date	2019-07-22
Active Health Index (0-100)	94.4%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for BF-CP-002:

The primary failure modes for Blast Furnace Cooling Pump #2 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

71. Asset Profile & FMEA: Hot Blast Blower (BF-BL-001)

Property	Specification Value
Asset Name / Tag ID	Hot Blast Blower / BF-BL-001
Plant Operational Area	Blast Furnace
Equipment Classification Type	Turbo Blower
Criticality Level	CRITICAL
Commissioning Install Date	2017-11-10
Active Health Index (0-100)	51.5%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for BF-BL-001:
The primary failure modes for Hot Blast Blower include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

72. Asset Profile & FMEA: BF Hydraulic System (BF-HY-001)

Property	Specification Value
Asset Name / Tag ID	BF Hydraulic System / BF-HY-001
Plant Operational Area	Blast Furnace
Equipment Classification Type	Hydraulic Power Unit
Criticality Level	HIGH
Commissioning Install Date	2020-01-05
Active Health Index (0-100)	93.3%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for BF-HY-001:

The primary failure modes for BF Hydraulic System include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

73. Asset Profile & FMEA: Raw Material Conveyor Belt (BF-CV-001)

Property	Specification Value
Asset Name / Tag ID	Raw Material Conveyor Belt / BF-CV-001
Plant Operational Area	Blast Furnace
Equipment Classification Type	Belt Conveyor
Criticality Level	HIGH
Commissioning Install Date	2021-06-18
Active Health Index (0-100)	93.7%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for BF-CV-001:

The primary failure modes for Raw Material Conveyor Belt include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

74. Asset Profile & FMEA: LD Converter Vessel #1 (SMS-LD-001)

Property	Specification Value
Asset Name / Tag ID	LD Converter Vessel #1 / SMS-LD-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	BOF Converter
Criticality Level	CRITICAL
Commissioning Install Date	2016-09-01
Active Health Index (0-100)	94.0%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SMS-LD-001:

The primary failure modes for LD Converter Vessel #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

75. Asset Profile & FMEA: Continuous Caster #1 (SMS-CC-001)

Property	Specification Value
Asset Name / Tag ID	Continuous Caster #1 / SMS-CC-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	Continuous Casting Machine
Criticality Level	CRITICAL
Commissioning Install Date	2018-12-20
Active Health Index (0-100)	50.7%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for SMS-CC-001:

The primary failure modes for Continuous Caster #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

76. Asset Profile & FMEA: Ladle Furnace (SMS-LF-001)

Property	Specification Value
Asset Name / Tag ID	Ladle Furnace / SMS-LF-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	Electric Arc Furnace
Criticality Level	HIGH
Commissioning Install Date	2019-04-14
Active Health Index (0-100)	94.8%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SMS-LF-001:

The primary failure modes for Ladle Furnace include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

77. Asset Profile & FMEA: EOT Crane #1 (250T) (SMS-CR-001)

Property	Specification Value
Asset Name / Tag ID	EOT Crane #1 (250T) / SMS-CR-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	Overhead Crane
Criticality Level	CRITICAL
Commissioning Install Date	2017-02-28
Active Health Index (0-100)	94.5%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SMS-CR-001:

The primary failure modes for EOT Crane #1 (250T) include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

78. Asset Profile & FMEA: Argon Stirring Pump (SMS-PU-001)

Property	Specification Value
Asset Name / Tag ID	Argon Stirring Pump / SMS-PU-001
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	Gas Compressor
Criticality Level	MEDIUM
Commissioning Install Date	2022-08-10
Active Health Index (0-100)	93.5%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SMS-PU-001:

The primary failure modes for Argon Stirring Pump include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

79. Asset Profile & FMEA: Rolling Mill Drive Motor (RM-DM-001)

Property	Specification Value
Asset Name / Tag ID	Rolling Mill Drive Motor / RM-DM-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	AC Motor (2000kW)
Criticality Level	CRITICAL
Commissioning Install Date	2018-05-30
Active Health Index (0-100)	84.3%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for RM-DM-001:

The primary failure modes for Rolling Mill Drive Motor include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

80. Asset Profile & FMEA: Mill Gearbox #1 (RM-GB-001)

Property	Specification Value
Asset Name / Tag ID	Mill Gearbox #1 / RM-GB-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	Helical Gearbox
Criticality Level	CRITICAL
Commissioning Install Date	2019-10-15
Active Health Index (0-100)	35.1%
Operational Status Status	CRITICAL
Downtime Risk Level Score	CRITICAL

Failure Mode and Effects Analysis (FMEA) for RM-GB-001:

The primary failure modes for Mill Gearbox #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

81. Asset Profile & FMEA: Roughing Stand (RM-RS-001)

Property	Specification Value
Asset Name / Tag ID	Roughing Stand / RM-RS-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	Rolling Stand
Criticality Level	HIGH
Commissioning Install Date	2020-03-22
Active Health Index (0-100)	94.2%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for RM-RS-001:

The primary failure modes for Roughing Stand include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

82. Asset Profile & FMEA: Finishing Stand #1 (RM-FS-001)

Property	Specification Value
Asset Name / Tag ID	Finishing Stand #1 / RM-FS-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	Rolling Stand
Criticality Level	HIGH
Commissioning Install Date	2020-03-22
Active Health Index (0-100)	94.1%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for RM-FS-001:

The primary failure modes for Finishing Stand #1 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

83. Asset Profile & FMEA: Cooling Bed System (RM-CL-001)

Property	Specification Value
Asset Name / Tag ID	Cooling Bed System / RM-CL-001
Plant Operational Area	Rolling Mill
Equipment Classification Type	Cooling System
Criticality Level	MEDIUM
Commissioning Install Date	2021-01-10
Active Health Index (0-100)	92.9%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for RM-CL-001:

The primary failure modes for Cooling Bed System include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

84. Asset Profile & FMEA: Coke Oven Pusher Machine (CO-PU-001)

Property	Specification Value
Asset Name / Tag ID	Coke Oven Pusher Machine / CO-PU-001
Plant Operational Area	Coke Oven
Equipment Classification Type	Pusher Machine
Criticality Level	HIGH
Commissioning Install Date	2019-08-05
Active Health Index (0-100)	88.9%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for CO-PU-001:

The primary failure modes for Coke Oven Pusher Machine include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Coke Oven area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

85. Asset Profile & FMEA: Quenching Car (CO-QC-001)

Property	Specification Value
Asset Name / Tag ID	Quenching Car / CO-QC-001
Plant Operational Area	Coke Oven
Equipment Classification Type	Quenching Car
Criticality Level	MEDIUM
Commissioning Install Date	2020-11-12
Active Health Index (0-100)	93.7%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for CO-QC-001:

The primary failure modes for Quenching Car include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Coke Oven area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

86. Asset Profile & FMEA: Gas Cleaning Plant (CO-GC-001)

Property	Specification Value
Asset Name / Tag ID	Gas Cleaning Plant / CO-GC-001
Plant Operational Area	Coke Oven
Equipment Classification Type	Gas Treatment System
Criticality Level	HIGH
Commissioning Install Date	2018-04-20
Active Health Index (0-100)	93.3%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for CO-GC-001:

The primary failure modes for Gas Cleaning Plant include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Coke Oven area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

87. Asset Profile & FMEA: Sinter Fan Main Blower (SP-FM-001)

Property	Specification Value
Asset Name / Tag ID	Sinter Fan Main Blower / SP-FM-001
Plant Operational Area	Sinter Plant
Equipment Classification Type	Centrifugal Fan
Criticality Level	CRITICAL
Commissioning Install Date	2017-06-15
Active Health Index (0-100)	49.1%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for SP-FM-001:

The primary failure modes for Sinter Fan Main Blower include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Sinter Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

88. Asset Profile & FMEA: Ignition Furnace (SP-IG-001)

Property	Specification Value
Asset Name / Tag ID	Ignition Furnace / SP-IG-001
Plant Operational Area	Sinter Plant
Equipment Classification Type	Gas Furnace
Criticality Level	HIGH
Commissioning Install Date	2019-09-30
Active Health Index (0-100)	93.6%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SP-IG-001:

The primary failure modes for Ignition Furnace include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Sinter Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

89. Asset Profile & FMEA: Sinter Mix Conveyor (SP-CV-001)

Property	Specification Value
Asset Name / Tag ID	Sinter Mix Conveyor / SP-CV-001
Plant Operational Area	Sinter Plant
Equipment Classification Type	Belt Conveyor
Criticality Level	MEDIUM
Commissioning Install Date	2021-02-14
Active Health Index (0-100)	93.6%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SP-CV-001:

The primary failure modes for Sinter Mix Conveyor include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Sinter Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

90. Asset Profile & FMEA: Steam Turbine Generator (PP-TG-001)

Property	Specification Value
Asset Name / Tag ID	Steam Turbine Generator / PP-TG-001
Plant Operational Area	Power Plant
Equipment Classification Type	Steam Turbine
Criticality Level	CRITICAL
Commissioning Install Date	2016-01-20
Active Health Index (0-100)	50.6%
Operational Status Status	DEGRADED
Downtime Risk Level Score	HIGH

Failure Mode and Effects Analysis (FMEA) for PP-TG-001:

The primary failure modes for Steam Turbine Generator include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Power Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

91. Asset Profile & FMEA: Boiler Feed Pump (PP-BL-001)

Property	Specification Value
Asset Name / Tag ID	Boiler Feed Pump / PP-BL-001
Plant Operational Area	Power Plant
Equipment Classification Type	Multi-stage Pump
Criticality Level	HIGH
Commissioning Install Date	2018-07-08
Active Health Index (0-100)	88.8%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for PP-BL-001:
The primary failure modes for Boiler Feed Pump include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Power Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

92. Asset Profile & FMEA: Cooling Tower Fan (PP-CT-001)

Property	Specification Value
Asset Name / Tag ID	Cooling Tower Fan / PP-CT-001
Plant Operational Area	Power Plant
Equipment Classification Type	Axial Fan
Criticality Level	MEDIUM
Commissioning Install Date	2020-05-25
Active Health Index (0-100)	93.7%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for PP-CT-001:
The primary failure modes for Cooling Tower Fan include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Power Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:
Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

93. Asset Profile & FMEA: Main Transformer (100MVA) (PP-TR-001)

Property	Specification Value
Asset Name / Tag ID	Main Transformer (100MVA) / PP-TR-001
Plant Operational Area	Power Plant
Equipment Classification Type	Power Transformer
Criticality Level	CRITICAL
Commissioning Install Date	2015-11-30
Active Health Index (0-100)	93.6%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for PP-TR-001:

The primary failure modes for Main Transformer (100MVA) include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Power Plant area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

94. Asset Profile & FMEA: Blast Furnace Cooling Pump #3 (BF-CP-003)

Property	Specification Value
Asset Name / Tag ID	Blast Furnace Cooling Pump #3 / BF-CP-003
Plant Operational Area	Blast Furnace
Equipment Classification Type	Centrifugal Pump
Criticality Level	CRITICAL
Commissioning Install Date	2026-08-10
Active Health Index (0-100)	100.0%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for BF-CP-003:

The primary failure modes for Blast Furnace Cooling Pump #3 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Blast Furnace area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

95. Asset Profile & FMEA: EOT Crane #3 (SMS-CR-003)

Property	Specification Value
Asset Name / Tag ID	EOT Crane #3 / SMS-CR-003
Plant Operational Area	Steel Melting Shop
Equipment Classification Type	EOT Crane
Criticality Level	HIGH
Commissioning Install Date	2026-09-01
Active Health Index (0-100)	100.0%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for SMS-CR-003:

The primary failure modes for EOT Crane #3 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Steel Melting Shop area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

96. Asset Profile & FMEA: Mill Gearbox #3 (RM-GB-003)

Property	Specification Value
Asset Name / Tag ID	Mill Gearbox #3 / RM-GB-003
Plant Operational Area	Rolling Mill
Equipment Classification Type	Gearbox
Criticality Level	CRITICAL
Commissioning Install Date	2026-08-25
Active Health Index (0-100)	100.0%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for RM-GB-003:

The primary failure modes for Mill Gearbox #3 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Rolling Mill area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

97. Asset Profile & FMEA: Steam Turbine #3 (PU-ST-003)

Property	Specification Value
Asset Name / Tag ID	Steam Turbine #3 / PU-ST-003
Plant Operational Area	Power Utility
Equipment Classification Type	Steam Turbine
Criticality Level	CRITICAL
Commissioning Install Date	2026-10-05
Active Health Index (0-100)	100.0%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for PU-ST-003:

The primary failure modes for Steam Turbine #3 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Power Utility area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

98. Asset Profile & FMEA: Extractor Fan #3 (CO-EX-003)

Property	Specification Value
Asset Name / Tag ID	Extractor Fan #3 / CO-EX-003
Plant Operational Area	Coke Oven
Equipment Classification Type	Extractor Fan
Criticality Level	HIGH
Commissioning Install Date	2026-09-15
Active Health Index (0-100)	100.0%
Operational Status Status	OPERATIONAL
Downtime Risk Level Score	LOW

Failure Mode and Effects Analysis (FMEA) for CO-EX-003:

The primary failure modes for Extractor Fan #3 include bearing wear, rotor misalignment, and mechanical seal decay. Under continuous load conditions in the Coke Oven area, excessive heat and vibration represent the initial signs of wear.

Maintenance Recommendation:

Check telemetry data weekly. If the isolation forest anomaly score drops below 60%, check alignment, inspect lubrication quality, and ensure the safety bypass is functional prior to inspection. Always follow the standard Lock Out Tag Out (LOTO) safety protocols.

99. Conclusion & Enterprise Roadmap

The **Maintenance Wizard** platform successfully integrates real-time telemetry analytics with multi-agent coordination. The system replaces legacy, siloed databases with an intelligent digital twin console, improving operational visibility and reducing unplanned downtime.

Future Enterprise Scaling Roadmap:

- **Edge Integration:** Run Isolation Forest models directly on edge telemetry collectors for sub-10ms anomaly detection.
- **AR Maintenance:** Integrate augmented reality headsets to display LOTO safety guides and parts inventory locations on physical machines.
- **Expanded FMEA Models:** Develop models for high-speed rolling mill reducers, converter tilt drives, and slag ladles.
- **Acoustic Analytics:** Train audio classification models to detect bearing wear and cavitation from microphone records.